De La Salle University

Fama-French 5-Factor Model Evaluation

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By:

ABRIL, Jehan
GO, Keira Ley Arcala
SISON, Aaron Joshua Estacio
VALENCIA, Carlos Martin Belangel

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1 Introduction

This document provides a rigorous evaluation of the **Fama-French 5-Factor Model** using R. The model extends the Capital Asset Pricing Model (CAPM) by incorporating Size, Value, Profitability, and Investment factors alongside standard market risk (Fama & French, 1993, 2015). The core objective is to determine the efficacy of these five factors in explaining the excess returns of a portfolio comprising eight selected assets. The shift from the single-factor CAPM was driven by its poor empirical record, which, as noted by Fama and French (2004), is often considered insufficient for practical applications. While the CAPM posits that risk is measured solely relative to a comprehensive market portfolio, empirical tests frequently show that the relation between average return and beta is flatter than predicted. This gap between theory and evidence necessitated a multidimensional approach to capture the risk factors that a single market beta could not account for.

1.1 Transition from CAPM to Fama-French 5-Factor Model

The Fama-French Model, developed by Eugene F. Fama and Kenneth R. French in September 1992 in their paper titled 'The Cross-Section of Expected Stock Return', was originally composed of three (3) factors and serves as an augmentation to the Capital Asset Pricing Model (CAPM). The primary purpose of the introduction of this model was to demonstrate that market risk alone cannot fully explain stock returns. Thus, the transition from a single-factor (beta) linear model (CAPM) to a multi-factor linear model led to the creation of a model that could provide a multidimensional assessment for expected returns. Here, the CAPM model was expanded to include firm-specific variables that are historically known to have an effect on returns.

The initial Fama-French Model was composed of the following systematic risk factors. First, Market Risk (Rf), taken from the original CAPM model, represents the excess returns of the overall market over the overall risk-free rate. Next, SMB (small minus big) represents the idea that small-cap stocks tend to outperform large-cap stocks over time. Lastly, there is HML (high minus low) or the value factor; stocks with higher BM ratios tend to outperform stocks with low BM ratios. The three-factor model was originally developed to encapsulate the relationship between average return and price ratios (such as book-to-market ratio) (Fama and French, 2015).

Moving forward, the Fama-French 3-Factor Model was further augmented by Fama and French in 2015 in their working paper titled 'A five-factor asset pricing model'. This model is composed of the three factors mentioned above with the further additions of RMW (robust minus weak) and CMA (conservative minus aggressive). The factor of RMW represents the difference between the returns of stocks with robust and weak profitability. Meanwhile, CMA represents the difference between the returns on diversified portfolios of stocks of low and high investment firms, which are referred to as conservative and aggressive (Fama and French, 2015). These labels refer to a company's investment policy, whether it can be considered conservative or aggressive (Mosoeu and Kodongo, 2022). This new model with these two newly added variables is what is now referred to as the Fama-French 5-Factor Model.

1.2 Theoretical Significance and Professional Relevance

The 5-factor model holds immense practical value for finance professionals, particularly in the realms of portfolio evaluation and cost of capital estimation. By identifying specific anomalies—such as the size and value effects—the model allows fund managers to more accurately assess whether a portfolio's "alpha" is the result of genuine skill or simply exposure to known risk premiums.

Furthermore, the model's ability to provide a more nuanced risk assessment assists analysts in estimating the cost of equity more reliably than the traditional CAPM. Understanding these factors is vital for asset pricing, as it helps investors recognize that expected returns are tied not just to market volatility, but to the underlying characteristics of the firms themselves, such as their profitability and investment policies (Fama & French, 2015).

1.3 Objectives

The primary goal of this research is to evaluate the explanatory power of the Fama-French 5-Factor Model through the following specific objectives:

- **Empirical Analysis:** To assess the efficacy of the five risk premiums—Market, Size (SMB), Value (HML), Profitability (RMW), and Investment (CMA)—in capturing the variations in returns across a diverse portfolio (Fama & French, 2015).
- **Statistical Interpretation:** To analyze the resulting alpha and beta coefficients for statistical significance (p-values) and interpret their economic implications for portfolio risk management.
- **Technical Implementation:** To develop and execute a fully reproducible R workflow that leverages the Kenneth French Data Library for factor ingestion and data alignment (French, 2024).
- **Theoretical Justification:** To provide rigorous academic grounding for the selection of eight specific assets, justifying their inclusion based on hypothesized factor exposures.

2 Methodology

2.1 Technical Workflow and Data Integration

The methodology for this study follows a systematic, reproducible workflow designed to align historical asset returns with the Fama-French risk premiums. The process is divided into data acquisition, preprocessing, and econometric modeling stages, ensuring that the results are statistically robust and grounded in established financial theory.

2.1.1 Data Acquisition and Ingestion

Two primary data streams are utilized for this analysis. Factor risk premiums—comprising the independent variables (Market, SMB, HML, RMW, and CMA) and the risk-free rate (R_f)—are sourced from the Kenneth R. French Data Library. We specifically utilize the "U.S. Research Returns" archive to maintain consistency with historical benchmarks. Asset return data, consisting of historical adjusted closing prices for the eight selected assets, are retrieved via the `tidyquant` and `quantmod` R packages. These prices are then transformed into monthly log-returns to facilitate linear regression modeling.

2.1.2 Preprocessing and Data Alignment

A critical component of the technical implementation is the synchronization of temporal indices. Since asset data and factor data often utilize different date formats, all indices are normalized to a standard YYYY-MM-01 format to ensure a perfect merge. Following standard asset pricing theory, we calculate the excess returns for each asset by subtracting the monthly risk-free rate (R_f) from the raw monthly returns ($R_i - R_f$). Furthermore, in accordance with the Kenneth French Library protocols, any missing values (typically coded as -99.99 or -99.9) are identified and handled to prevent skewing the regression results.

2.1.3 Multivariate Regression Framework

The core analytics are performed using Ordinary Least Squares (OLS) regression within the R environment. We estimate the following 5-factor equation for each of our eight selected assets:

$$R_{it} - R_{ft} = \alpha_i + \beta_{i1}(R_{Mt} - R_{ft}) + \beta_{i2}SMB_t + \beta_{i3}HML_t + \beta_{i4}RMW_t + \beta_{i5}CMA_t + \epsilon_{it} \quad (1)$$

This model allows us to decompose the source of returns for our portfolio, identifying whether performance is driven by market risk, size, value, profitability, or investment intensity.

2.2 Asset Selection

The portfolio analyzed in this study consists of eight diverse assets chosen for their hypothesized exposures to various risk factors: VUSA.L, WSML.L, VEVE.L, IWQU.L, VIG, IITU.L, SGLN.L, and VWRA.L. These assets represent various tilts toward Market, SMB, HML, RMW, and CMA factors, providing a comprehensive dataset for evaluating the model's explanatory power across different asset classes and investment styles. Table 1 details the specific academic and technical justification for each selection.

To capture the specific risk premiums identified by the Fama-French framework, the portfolio includes a selection of eight exchange-traded funds (ETFs) with distinct factor tilts. The baseline for market exposure is established through **VUSA.L** (Vanguard S&P 500 UCITS), which serves as the primary proxy for broad U.S. equity market beta ($R_i - R_f$). This is complemented by **VWRA.L** (Vanguard FTSE All-World), representing the global market portfolio and ensuring the model accounts for aggregate risk across both developed and emerging regions. To further refine geographical exposure, **VEVE.L** (Vanguard FTSE

Table 1: Asset Selection and Factor Exposure

Asset Ticker	**Factor Exposure**
VUSA.L	Market (Beta)
WSML.L	Size (SMB)
VEVE.L	Market / Developed
IWQU.L	Profitability (RMW)
VIG	Investment (CMA)
IITU.L	Growth (Low HML)
SGLN.L	Diversification
VWRA.L	Total Market

Developed World) is utilized to test the model's validity across the broader developed-market universe beyond the United States.

Factor-specific tilts are achieved through targeted allocations. The size premium is captured via **WSML.L** (iShares MSCI World Small Cap), which holds a diversified set of global small-capitalization firms hypothesized to outperform large-cap counterparts over time. The profitability (RMW) factor is represented by **IWQU.L** (iShares Edge MSCI World Quality), which filters for companies with high operating profitability and stable earnings. Conversely, the investment (CMA) factor is addressed through **VIG** (Vanguard Dividend Appreciation); firms with consistent dividend increases typically follow conservative reinvestment policies, prioritizing shareholder returns over aggressive expansion.

To provide a counter-balance to value-oriented factors, **IITU.L** (iShares S&P 500 Info Tech) introduces a significant growth tilt, characterized by lower book-to-market ratios. Finally, **SGLN.L** (iShares Physical Gold) is included as a non-equity stabilizer. While gold does not carry direct exposure to standard Fama-French equity premiums, its inclusion serves a critical diagnostic purpose: it allows for an evaluation of the model's intercept behavior (Alpha) when applied to assets with zero hypothesized factor loadings.

2.3 Descriptive Statistics

Historical return data for the selected assets is analyzed using mean, median, standard deviation, skewness, and kurtosis to understand the distribution and risk profile of the portfolio.

3 Results

[Detailed results, regression coefficients, and visual analysis to be imported from Calculations.Rmd]

4 Conclusion

The study concludes with an academic verdict on the model's efficacy for the selected portfolio, linking empirical findings to practical investment implications.

References

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